3 **(a)** A block of mass 0.40 kg slides in a straight line with a constant speed of 0.30 m s^{-1} along a smooth horizontal surface, as shown in Fig. 3.1.

Fig. 3.1

The block hits a spring and decelerates. The speed of the block becomes zero when the spring is compressed by 8.0 cm.

(i) Calculate the initial kinetic energy of the block.

kinetic energy = ... J [1]

(ii) The variation of the compression *x* of the spring with the force *F* applied to the spring is shown in Fig. 3.2.

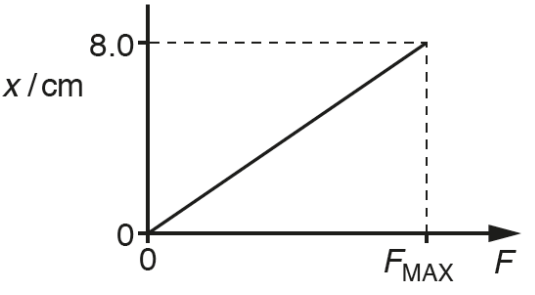

Fig. 3.2

Use your answer in **(a)(i)** to determine the maximum force F_{MAX} exerted on the spring by the block. Explain your working.

F_{MAX} = ... N [2]

(iii) Calculate the maximum deceleration of the block.

deceleration = ... m s^{-2} [1]

(iv) State and explain whether the block is in equilibrium

1. before it hits the spring,

.....

.....

2. when its speed becomes zero.

.....

..... [2]

(b) The energy E stored in a spring is given by

$$E = \frac{1}{2}kx^2$$

where k is the spring constant of the spring and x is its compression.

The mass m of the block in **(a)** is now varied. The initial speed of the block remains constant and the spring continues to obey Hooke's law.

On Fig. 3.3, sketch the variation with mass m of the maximum compression x_0 of the spring.

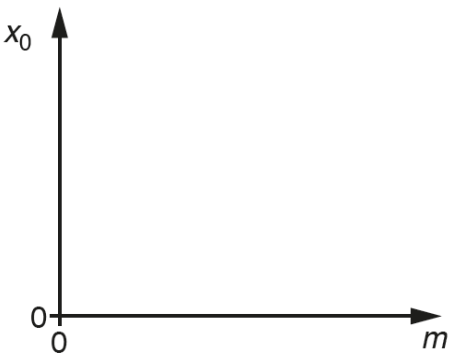

Fig. 3.3

[2]